



**VietNam National University
University of Engineering and Technology**

EMBEDDED SYSTEM FUNDAMENTALS

(ELT3240, NHẬP MÔN HỆ THỐNG NHÚNG)

Dr. Nguyễn Kiêm Hùng

Email: kiemhung@vnu.edu.vn

Laboratory for Smart Integrated Systems

Objectives

In this lecture you will be introduced to:

- Basic concepts of Embedded Systems
- What is **difference and unique** about embedding computing system and general-purpose computing system,

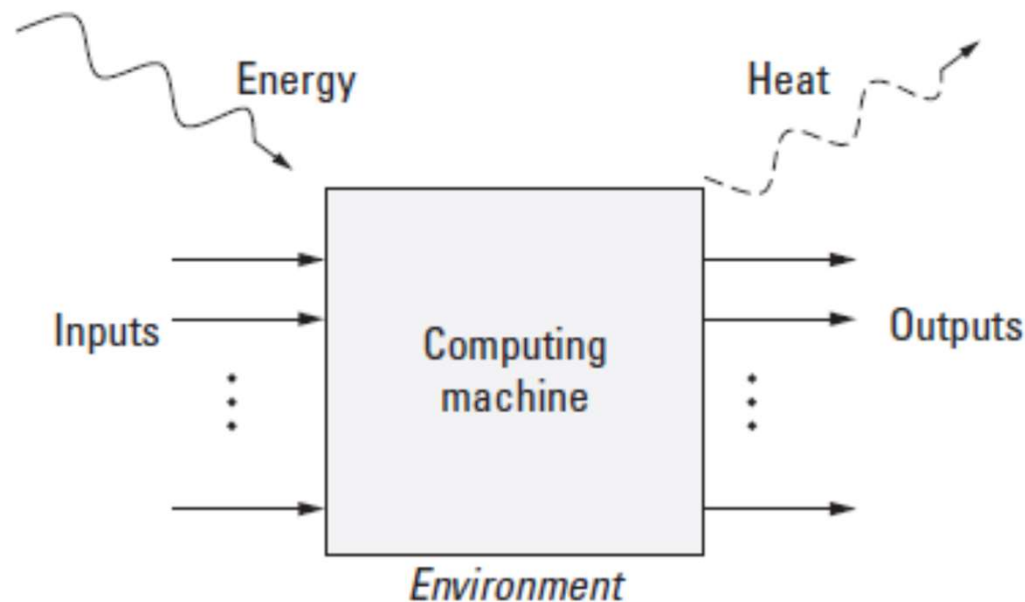
Outline



- **Embedded Computing System**
- Characteristics of Embedded Systems
- Challenges in Embedded System Design
- Summary

Review of Computing System

- A computing system (or just computer) is frequently modeled as a system that includes:
 - inputs: physical signals from the environment ,
 - outputs: response of system to the environment ,
 - and a **processing unit: fetches and executes instructions from a memory**

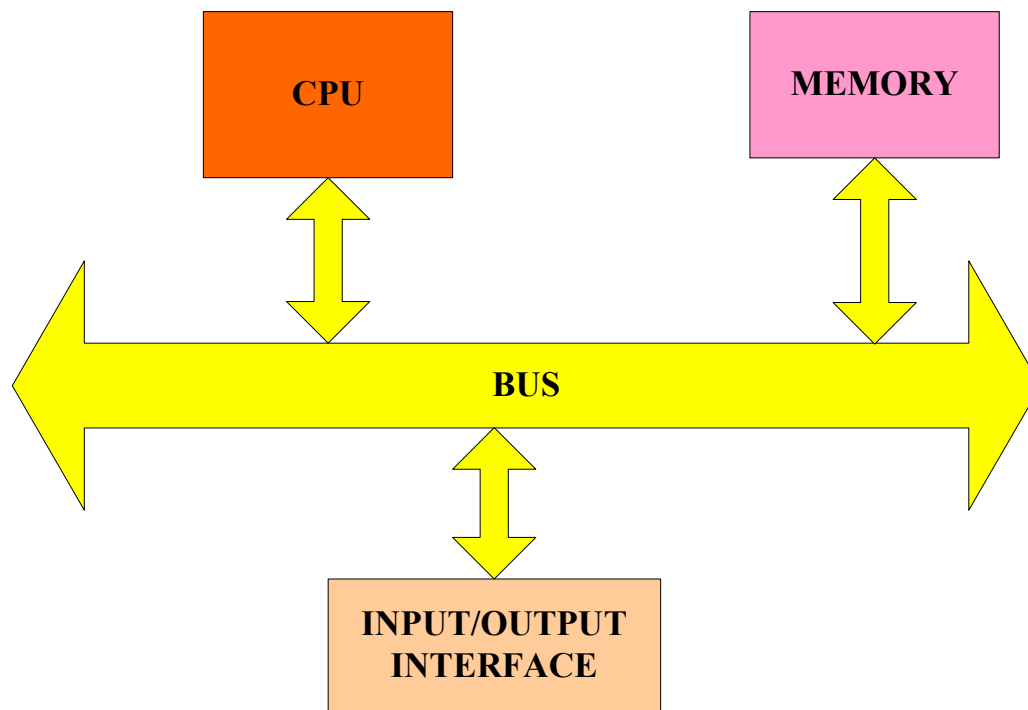


An abstract view of a computing system

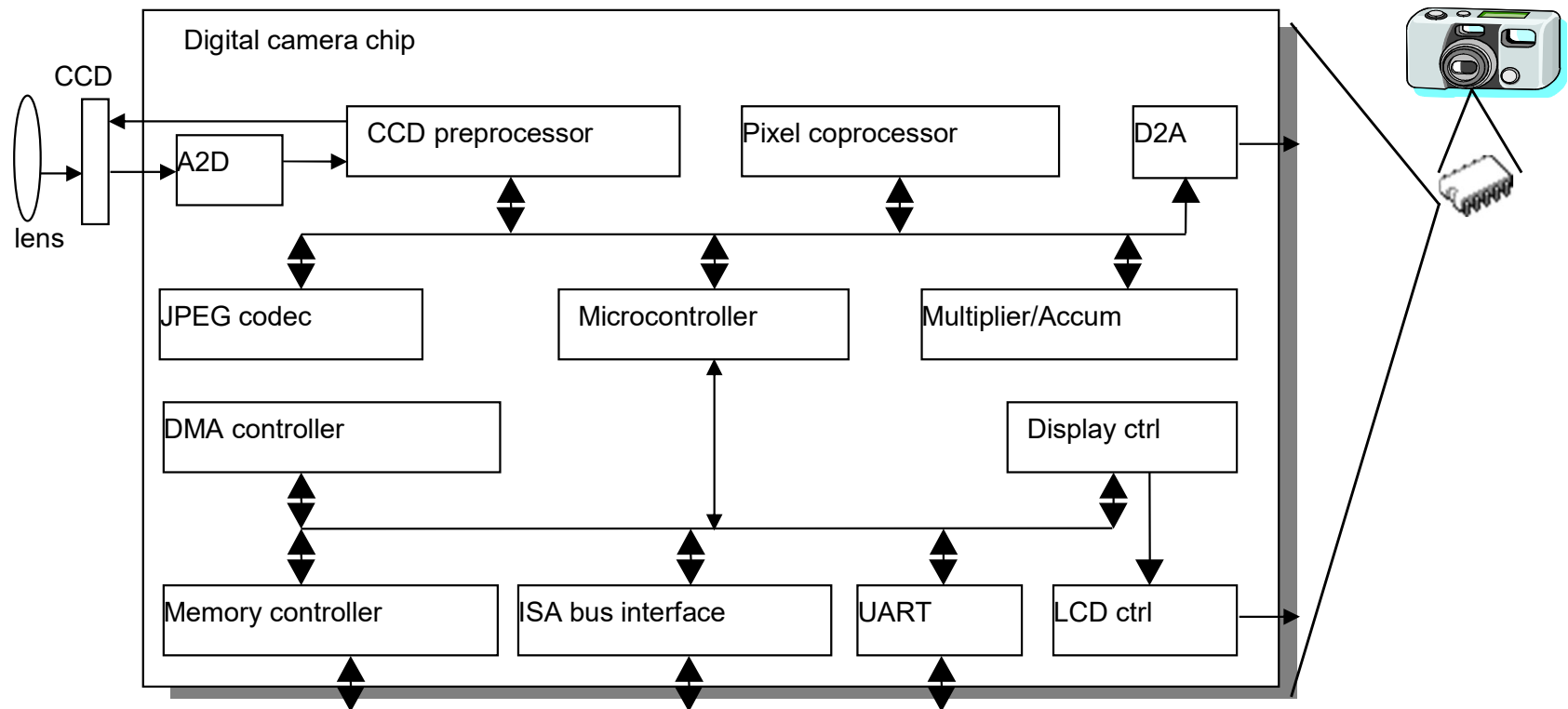
Review of Computing System



Which component does a minimal computing system consists of ?



An embedded system example -- a digital camera



- Single-functioned -- always a digital camera
- Tightly-constrained -- Low cost, low power, small, fast
- Reactive and real-time -- only to a small extent

Embedded Computing System

- An embedded system is a **specialized computing machine** (other than a general-purpose computer) with the following features:
 - Single dedicated function
 - Typically designed to perform a predefined function
 - Tightly constraints:
 - Low cost
 - Single-to-fewer component
 - Performs functions fast enough
 - Consumes minimum power
 - Real-time Response
 - Must continually monitor the desired environment and react to changes
 - Hardware and software coexistence
 - The program written for embedded system is called “firm-ware”, and usually stored in persistence memory (ROM/Flash)
 - The firm-ware can run with limited hardware resourced.

Embedded Computing System

- Embedded systems can be classified based on their complexity and performance into:
 - **Small-scale systems**
 - Perform simple functions, operate on battery, no operating system
 - usually built around low-end 8- or 16-bit microprocessors or microcontrollers
 - main programming tools: an editor, assembler, cross-assembler, and integrated development environment (IDE).
 - Examples: mouse and TV remote control
 - **Medium-scale systems**
 - have both hardware and software complexities, have operating system support
 - use 16- or 32-bit microprocessors or microcontrollers
 - main programming tools are C, C++, JAVA, Visual C++, debugger, source-code engineering tool, simulator, and IDE
 - Examples: vending machines and washing machines
 - **Large-scale or sophisticated systems**
 - have enormous hardware and software complexities
 - built around 32- or 64-bit microprocessors or microcontrollers, along with a range of other high-speed integrated circuits.
 - need hardware and software co-design techniques
 - Examples: flight-landing gear systems, car braking systems, and military applications

Embedded Computer vs. General-purpose Computer

How the computing machine is used?

Embedded System	General-purpose computer
- A component of some larger product; its purpose is narrowly focused on supporting that product	- A complete product itself
- End user of the product typically does not directly interact with the embedded system, or interacts with only a limited interface	- End user directly interacts with it
- User don't consider the product (e.g. DVD players, MP3 players, game consoles) they are buying is a computer	- User explicitly knows the product they are buying is a computer

Embedded Computer vs. General-purpose Computer

How the computing machine is used?

Embedded System

- Having some of these standard peripherals as well, but also include much more specialized ones:
 - + special-purpose sensors: accelerometers, temperature probes, magnetometers, push button contact switches, etc.
 - + special-purpose outputs: lamps and LEDs, actuators, TTL electrical signals, LCD displays, etc.

General-purpose computer

- Having relatively few, standardized inputs and outputs: keyboards, mice, network connections, video monitors, and printers

Embedded System Definition

“Embedded systems are computing systems with tightly coupled hardware and software integration, that are designed to perform a dedicated function. The word ‘embedded’ reflects the fact that these systems are usually an integral part of a larger (mechanical or electrical) system/product, known as the embedding system. Multiple embedded systems can coexist in an embedding system.” – (Qing Li and Carolyn Yao)

Embedded Computer vs. General-purpose Computer



Is a handheld computer an embedded system?

- Characteristics of general-purpose computer:
 - what is its enclosing product?
 - It is not used to control anything
 - the computing system is exposed to the user: the user can download general-purpose applications.
- Characteristics of embedded systems:
 - very sensitive to size, weight, and power constraints.
 - a limited user interface: no standardized inputs and outputs, but include much more specialized ones
 - Etc.

Embedded Computer vs. General-purpose Computer



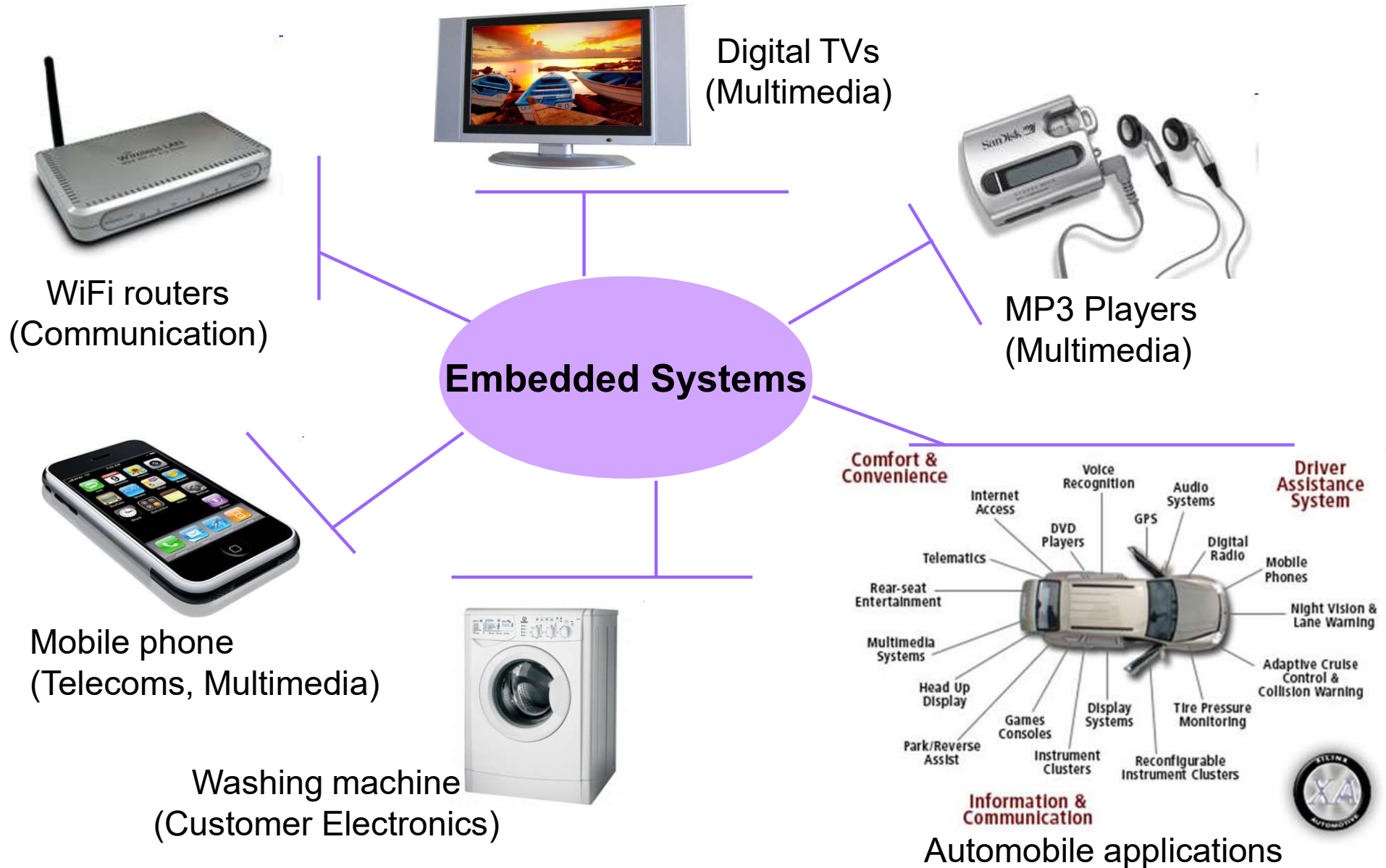
The exact boundary between general-purpose and embedded is not black and white but rather a spectrum.

Outline



- Embedded Computing System
- **Characteristics of Embedded Systems**
- Challenges in Embedded System Design
- Summary

Embedded Applications

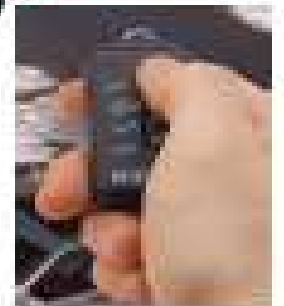


Laboratory for Smart Integrated Systems

Embedded Applications



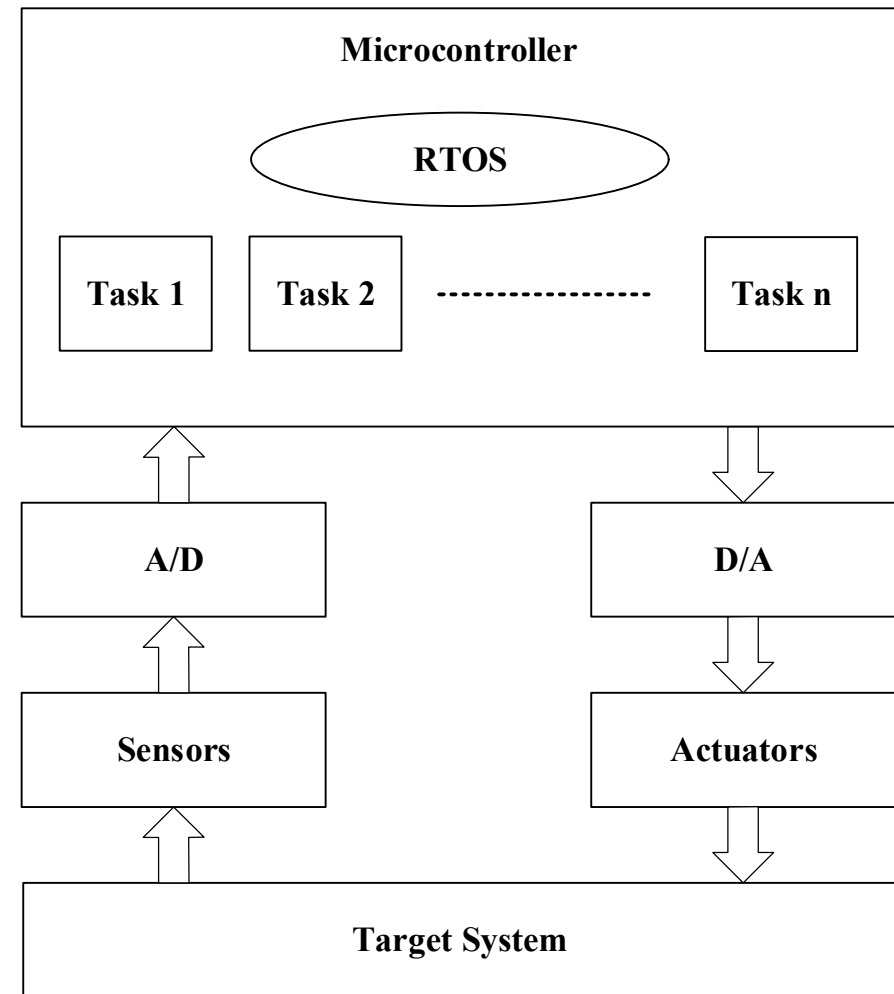
Embedded System =
Computers Inside a Product



Laboratory for Smart Integrated Systems

Characteristics of Embedded Systems

- System structure:
 - **Sensors:** to retrieve data from its environment
 - **ADC:** converts the analog data from the sensor to digital signals
 - **Controller:** the brain of the system; execute a set of real-time tasks concurrently, may or may not be with the support of a RTOS
 - **Actuators:** through actuators the controller acts upon the target system
 - **DAC:** convert the digital signals from microcontroller to the analog signal



Characteristics of Embedded Systems

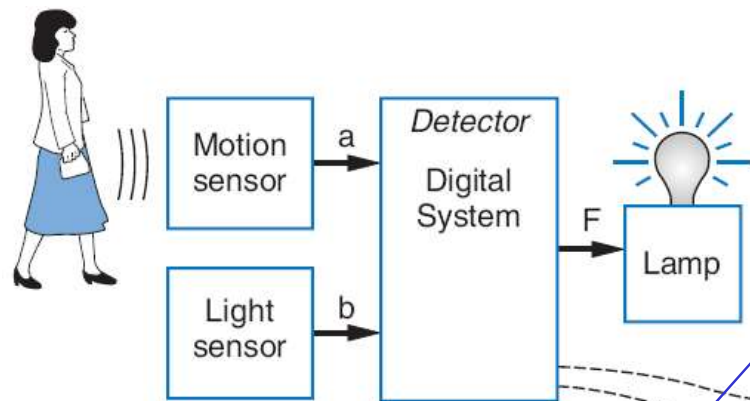
- Sophisticated functionality:
 - *Complex algorithms*: automobile, nuclear power, aerospace,
 - *User interface*: voice, image, ...
- Performance Constrains:
 - *Real time*:
 - Hard real-time
 - Soft real-time
 - *Multi-rate*
- Cost Requirements:
 - *Manufacturing cost*
 - *Power and energy*

Characteristics of Embedded Systems

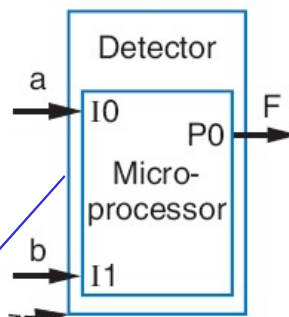
- **Highly Constrained Environments**
 - highly resource-constrained environments: processor speed, memory capacity, and user interface
- **Concurrency:**
 - several computations are executing simultaneously and potentially interacting with each other
- **Safety and Reliability**
 - Safety means “freedom from accidents or losses” in the absence of faults as well as in the presence of single-point faults.
 - Reliability refers to the ability of a system or component to perform its required functions under stated conditions for a specified time.
 - measured in failures per million operating hours

Implementing Embedded Systems: Programming Microprocessors Vs. Designing Digital Circuits

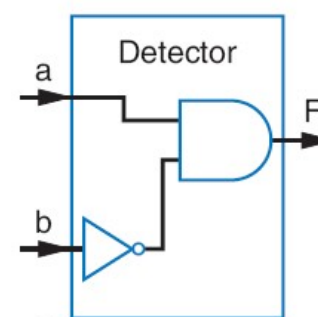
Desired motion-at-night detector



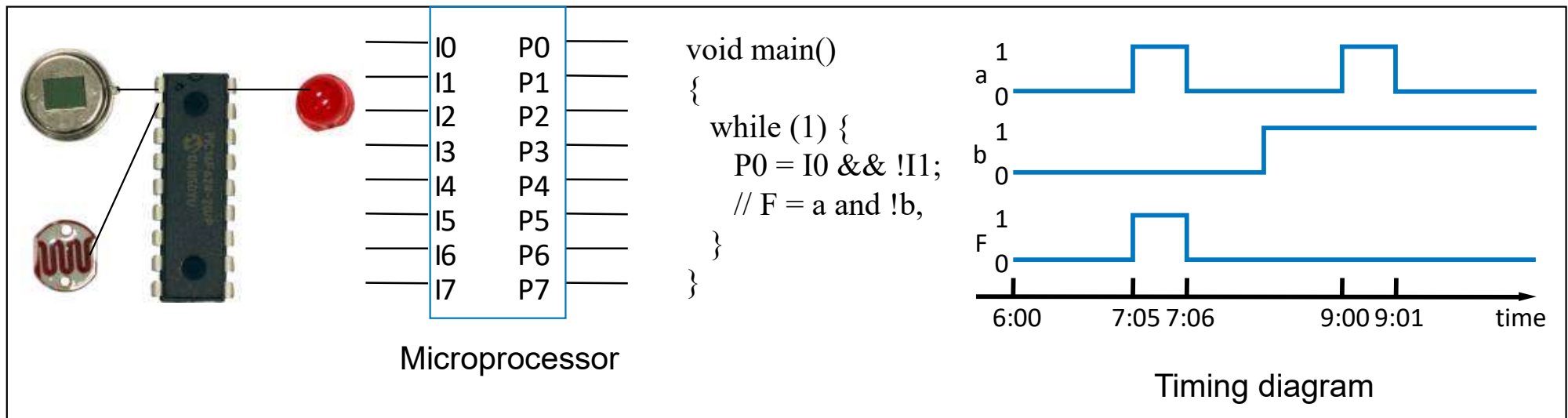
*Programmed
microprocessor*



*Custom designed
digital circuit*



- Microprocessors is a common choice to implement a digital system
 - Easy to program
 - Cheap (as low as \$1)
 - Available now

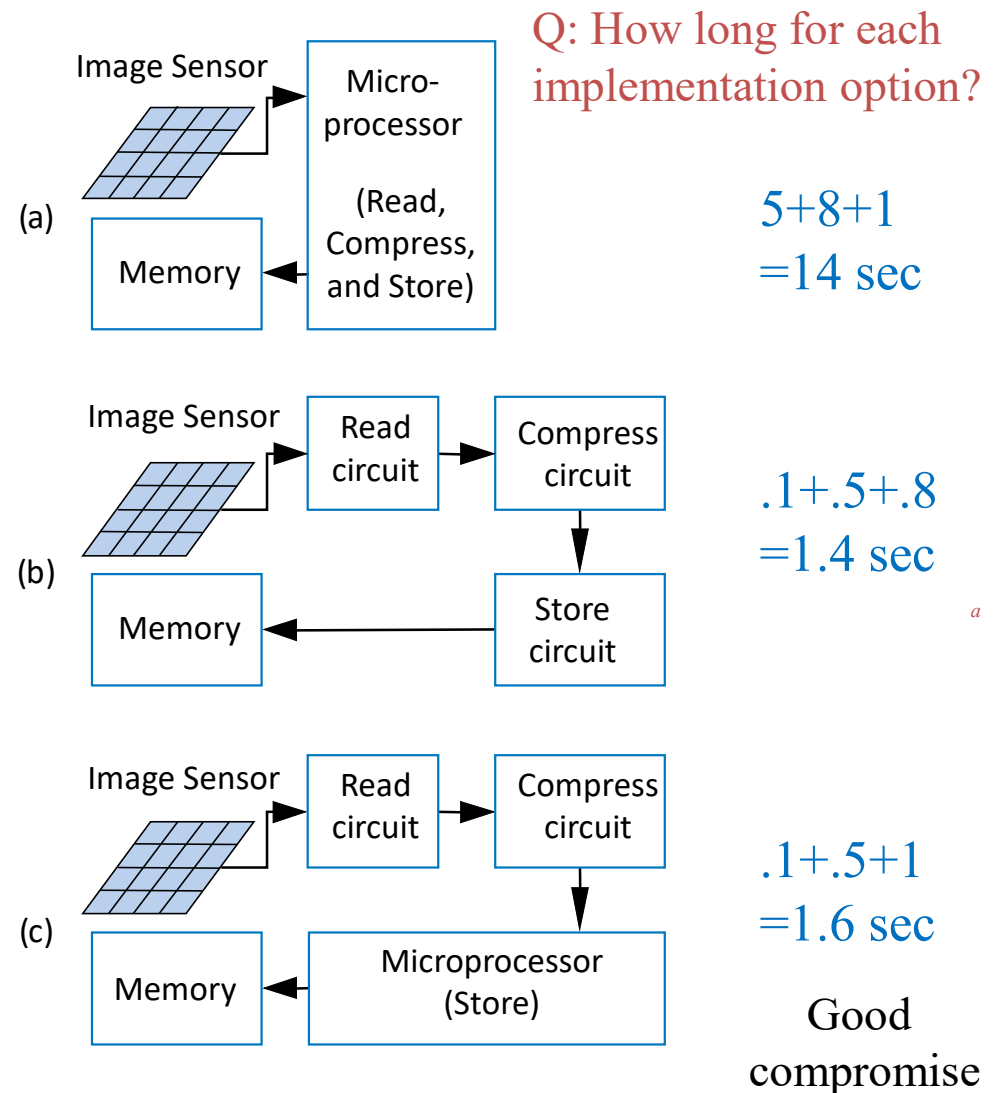


Embedded Systems Design: When Microprocessors Aren't Good Enough

- With microprocessors so easy, cheap, and available, why design a digital circuit?
 - Microprocessor may be too slow
 - Or too big, power hungry, or costly

Sample digital camera task execution times (in seconds) on a microprocessor versus a digital circuit:

Task	Microprocessor	Custom Digital Circuit
Read	5	0.1
Compress	8	0.5
Store	1	0.8



Why we embed microprocessor in system?

- **Microprocessors are a very efficient way to implement digital systems:**
 - high-performance processors can execute several instructions per cycle
 - CPUs are highly optimized for speed the latest manufacturing technology
 - A microprocessor, on the other hand, can be used for many different algorithms simply by changing the program it executes.
 - Implementation of your application is faster than designing your own custom logic
- **Microprocessors are flexible thank to their programmability:**
 - it is easier to design families of products
 - it is easier to provide new features to keep up with rapidly changing markets
 - it is possible to reuse software for next-generation products, thereby reducing development time and cost.

Why we embed microprocessor in system?



Why not use PCs for all embedded computing?

- **It is difficult to meet real-time performance requirements**
 - Real-time performance is often best achieved by RTOS based on priority-driven scheduling algorithms
- **It is difficult to meet Low power and low cost requirements:**
 - PCs are designed to satisfy a broad mix of computing requirements .
 - In contrast, custom embedded systems that are designed for an application.

Why we embed microprocessor in system?



Whether we can use Smartphones as platforms for embedded computing?

- Features:
 - Smartphones are very power efficient
 - Smartphones can perform huge amounts of computation in real time
 - Small Size
 - Popular

Outline

- Embedded Computing System
- Characteristics of Embedded Systems
- **Challenges in Embedded System Design**
- Summary



Challenges in Embedded System Design

- **Some important problems that must be taken into account in embedded system design:**
 - *How much hardware do we need?*
 - the choice of hardware must meet both **performance deadlines** and **manufacturing cost constraints**
 - *How do we meet time constraints?*
 - to speed up the hardware so that the program runs faster
 - *How do we minimize power consumption?*
 - slowing down the system
 - *How do we design for upgradeability?*
 - The design must be enough flexible for several product generations, or for several different versions of a product in the same generation
 - *How does it really work?*
 - Reliability is always important when selling products

Challenges in Embedded System Design

- **The factors makes embedded system designs more difficult**
 - *Complex testing:*
 - have to run a real machine in order to generate the proper data
 - *Limited observability and controllability:*
 - do not come with keyboards and screens
 - in real-time applications it is not easy to stop the system
 - *Restricted development environments:*
 - the tools used to develop software and hardware) are often much more limited

Outline

- Embedded Computing System
- Characteristics of Embedded Systems
- Challenges in Embedded System Design



- **Summary**

Summary

- **What we Learned**

- Basic concepts of Embedded Computing Systems
- Distinguish between embedded Computing Systems and general-purpose computing Systems

Review Questions

- Q1: What is an embedded computer system? What is difficult and unique about embedding computing and general-purpose computer?
- Q2: Name three consumer electronics products that have embedded systems. Name three consumer electronics products that do not contain embedded computer systems.